Time & Work – Formula Sheet

◆ 1. Basic Formula

Work = Time × Efficiency

◆ 2. Efficiency

If A completes work in n days:
Efficiency of A = 1 / n

◆ 3. Combined Work

If A takes x days and B takes y days:
(A + B) one day work = (1/x + 1/y)

Total time = (x × y) / (x + y)

◆ 4. Three Persons

1 day work = (1/x + 1/y + 1/z)

◆ 5. Work Ratio

If A : B = x : y
Then time ratio = y : x

◆ 6. Men, Days, Work

Work = Men × Days × Hours

◆ 7. LCM Method (Most Important)

Example:
A = 10 days, B = 15 days

Take LCM = 30 units

Efficiency:
A = 30/10 = 3 units/day
B = 30/15 = 2 units/day

◆ 8. Alternate Day Work

A works Day 1, B Day 2

Find work done in 2 days:
(A + B work)

Then repeat cycle

◆ 9. Pipes & Cisterns

Filling → Positive
Emptying → Negative

Net work = Inlet − Outlet

◆ 10. Work Done

Work done in t days = t / total days

Remaining work = 1 − (t / total days)

◆ 11. Efficiency Increase

If efficiency increases by x%

New time = Old time / (1 + x/100)

◆ 12. Work Distribution

If total work = W and ratio = a : b

A’s share = (a / (a + b)) × W
B’s share = (b / (a + b)) × W

◆ 13. Quick Trick

If A is 2 times efficient as B:
Time of A = Time of B / 2